

### Problem 3.14

Assume that  $A(t) = 100 + 5t$ , where  $t$  is in years.

- (a) Find the principal.
- (b) How much is the investment worth after 5 years?
- (c) How much is earned on this investment during the 5th year?

### Solution

- (a) The principal is the value at time 0:

$$A(0) = 100 + 5(0) = 100.$$

- (b) After 5 years,

$$A(5) = 100 + 5(5) = 125.$$

- (c) First compute the value after 4 years:

$$A(4) = 100 + 5(4) = 120.$$

Therefore, the amount earned during the 5th year is

$$I_5 = A(5) - A(4) = 125 - 120 = 5.$$